American Computer Science League

2022-2023 ● Contest 2: Short Problems ● Classroom Division

1. Prefix/Infix/Postfix Notation

Evaluate the following prefix expression if all numbers are single digits:

+ − * 2 3 / 4 2 ^ 2 3

A. 14
B. 4
C. 12
D. 10
E. 6

2. Prefix/Infix/Postfix Notation

Convert the following infix expression to postfix:

$$\frac{A+B}{C} - \frac{B*C}{A-C}$$

A. A B + C / B C * A C − / −
B. A + B C / B C * A C − / −
C. A B C + / B C * A C − / −
D. A B + C / B * C A C − / −
E. A B + C / B C * A − C / −

3. Bit-String Flicking

Evaluate:

10100 OR 01100 AND 01011

A. 01000
B. 11100
C. 00111
D. 11111
E. 00000

4. Bit-String Flicking

Evaluate:

(LCIRC-2 (RSHIFT-1 (NOT 01110)))

A. 00010
B. 10001
C. 00001
D. 00100
E. 01100

5. LISP

Evaluate the following sequence of LISP expressions:

(SETQ M '(1 2 (5 1) (3 (4 5) 6) (2 (4))))
(CDR (CAR (CDR (REVERSE (CDR (CDR M))))))

A. (5 1)
B. (4 5)
C. ((4 5) 6)
D. (3 (4 5))
E. (2 (4))

6. Prefix/Infix/Postfix Notation

Evaluate the following postfix expression if all numbers are single digits:

6 3 / 2 2 ^ + 3 4 2 – * /

A. 0
B. 1
C. 6
D. 12
E. 15

7. Prefix/Infix/Postfix Notation

Translate the following postfix expression into prefix if all numbers are single digits:

4 6 + 7 5 – / 2 ^ 7 5 4 – * +

A. +^/4+6–752*7–54
B. +^/+46–752*75–4
C. +^+/46–752*7–54
D. +^/+467–52*7–54
E. +^/+46–752*7–54

8. Bit-String Flicking

Evaluate the following expression:

(LSHIFT-2 (RCIRC-1 (NOT 10110)))

A. 01001
B. 10000
C. 00101
D. 10010
E. 01100

9. Bit-String Flicking

Solve the following expression for X if X is a 5-bit string:

(NOT 11011) AND X OR (RCIRC-1 01110) = 00111

A. 001*0
B. *0111
C. 00*00
D. *****
E. 01010

10. LISP

Evaluate the following LISP expression:

(CDR (CAR (CDR '(b (c (a) b)))))

A. (a b)
B. (c (a))
C. ((a) b)
D. (a)
E. (c (a) b)